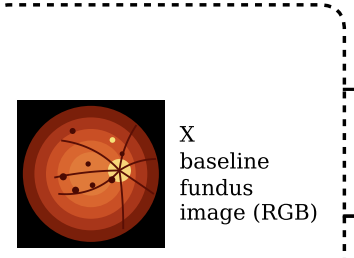
Module 1 - severity grading & lesion segmentation (DRG-Net)



X
baseline
fundus
image (RGB)

512 × 512, FGADR
normalisation

ResNet-50

FC 2048 → 5
softmax

ĝ grade
ICDR 0–4 +
class probabilities

to Module 2:
start stage ĝ, mask M
to Module 3:
ĝ, LBS stratum

U-Net × 4 (ResNet-50 encoder,
2-class output), one per lesion

MA

HE

EX

SE

∪

M
lesion
mask
(1 ch)

per-grade thresholds

Training (notebook 11)

Grading: FGADR
weighted cross-entropy
SGD, lr 1e-4, 20 ep

Segmentation:
FGADR + DDR combined
train 240 + 383 images
eval 81 + 149 images
pixel cross-entropy, 100 ep
best epoch = highest eval AP

FOV mask
(green channel > 10)

LBS =
|M| / |FOV|

burden stratum
low / med / high
(per-grade tertiles)